

MANGO, Togo

WMO: 653520

Lat: 10.367N

Lon: 0.467E

Elev: 146

StdP: 99.58

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	18.8	19.8	-3.7	2.8	27.7	-2.3	3.2	27.7	5.3	27.1	4.8	27.3	2.1	N/A	0.336

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	11.4	39.9	23.4	39.0	23.5	38.2	23.5	28.1	33.7	27.6	33.4	27.2	33.1	1.4	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.8	22.8	30.8	26.2	22.0	30.3	25.8	21.5	30.0	90.9	33.6	88.8	33.7	87.0	33.2	32.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.6	4.3	3.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			9.0	29.6	0.9	0.9	8.3	30.2	7.8	30.7	7.3	31.2	6.7	31.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.9	27.3	30.5	32.8	32.0	30.1	28.4	27.1	26.6	27.1	28.4	28.8	27.4
	DBStd	2.50	1.87	1.86	1.56	1.88	1.90	1.72	1.38	1.24	1.37	1.09	1.22	1.51
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6887	537	575	706	661	622	553	529	516	514	571	564	540
	CDD18.3	3845	279	342	447	411	363	303	271	257	264	312	314	282
	CDH23.3	49324	3548	5143	7159	6351	5068	3701	2723	2363	2554	3396	3907	3413
	CDH26.7	26708	1944	3181	4738	4029	2788	1714	977	745	934	1578	2220	1861
Wind	WSAvg	1.3	2.0	1.6	1.3	1.7	1.4	1.3	1.1	1.0	0.8	0.8	1.1	1.9
Precipitation	PrecAvg	1089	1	3	21	62	104	145	205	237	207	74	3	3
	PrecMax	1580	27	37	120	209	231	262	418	401	352	191	44	47
	PrecMin	902	0	0	0	2	25	54	84	94	69	2	0	0
	PrecStd	143	4	8	24	46	40	52	83	71	72	44	9	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.7	40.7	41.1	40.4	38.2	36.2	33.8	32.9	34.1	36.1	38.0	38.0
		MCWB	20.2	22.9	22.9	24.8	26.7	26.2	26.5	26.2	27.0	25.7	23.3	20.8
	2%	DB	36.8	39.1	40.0	39.5	37.0	35.0	32.7	31.9	32.8	35.0	37.0	36.6
		MCWB	19.7	22.1	23.1	25.0	26.6	26.3	26.4	26.2	26.6	26.0	23.2	20.7
	5%	DB	35.5	38.0	39.2	38.5	36.1	33.9	31.9	31.0	31.9	34.0	36.1	35.5
		MCWB	18.6	21.2	23.2	25.6	26.4	26.4	26.2	26.1	26.6	26.1	23.3	19.8
	10%	DB	34.2	37.1	38.5	37.6	35.2	33.0	31.0	30.2	31.0	32.9	35.2	34.2
		MCWB	17.9	20.7	23.3	25.9	26.5	26.3	26.0	26.0	26.3	26.2	23.2	19.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.1	26.3	27.7	29.3	28.6	28.5	27.9	27.4	28.1	28.2	27.2	25.8
		MCDB	34.8	37.1	37.1	35.5	34.2	32.8	31.4	31.1	31.9	32.1	34.6	34.6
	2%	WB	22.0	25.2	26.7	27.7	27.7	27.6	27.0	26.8	27.5	27.6	26.3	24.3
		MCDB	33.3	36.0	36.2	35.6	34.2	32.7	31.2	30.5	31.5	32.0	34.1	33.1
	5%	WB	20.1	24.3	26.1	27.2	27.3	27.0	26.6	26.4	27.0	27.2	25.7	22.5
		MCDB	33.6	34.2	35.8	35.3	33.9	32.1	30.6	30.0	31.0	31.9	33.4	31.4
	10%	WB	18.9	23.4	25.5	26.8	26.9	26.6	26.2	26.1	26.5	26.7	25.0	20.6
		MCDB	33.0	33.3	35.0	35.0	33.6	31.6	30.1	29.5	30.3	31.5	32.4	32.5
Mean Daily Temperature Range	5% DB	MDBR	13.9	13.1	11.4	10.0	8.8	8.2	7.0	6.6	7.6	10.1	14.8	14.8
		MCDBR	16.1	14.3	12.7	11.4	10.2	9.7	8.7	8.2	9.2	11.6	15.6	16.6
		MCWBR	7.1	6.5	5.0	3.6	3.0	3.1	3.1	3.1	3.9	4.1	5.2	6.9
	5% WB	MCDBR	15.0	12.2	10.8	9.9	9.3	8.7	8.0	7.4	8.7	10.0	13.5	15.2
		MCWBR	7.9	6.5	4.2	3.5	3.5	3.5	3.4	3.3	4.1	4.3	5.1	7.9
Clear-Sky Solar Irradiance	taub	0.679	0.742	0.812	0.721	0.631	0.579	0.528	0.514	0.521	0.605	0.599	0.598	
	taud	1.635	1.535	1.450	1.655	1.790	1.924	2.103	2.145	2.100	1.856	1.805	1.788	
	Ebn at Noon	659	638	605	658	705	734	776	799	801	728	716	709	
	Edh at Noon	255	290	320	259	221	191	160	156	165	208	213	214	
All-Sky Solar Radiation	RadAvg	5.42	5.62	5.82	5.97	5.80	5.31	4.79	4.58	4.99	5.67	5.57	5.33	
	RadStd	0.23	0.21	0.26	0.21	0.20	0.16	0.17	0.20	0.19	0.18	0.12	0.19	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page